The Role of Credit Constraint in Collateral Channel of Emerging Economies*

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March 18, 2025

Abstract

This paper examines the role of the credit constraint as a critical component in the effectiveness of collateral channel as a mechanism of monetary policy in emerging market economies (EMEs). By decomposing credit constraints, we developed an adjusted Permanent Income Hypothesis (PIH) model, incorporating economic uncertainty, loan-to-value (LTV) limits, financial development levels, and asset liquidation cost ratio to better capture collateralized consumption behaviour in EMEs. We argue that low financial development and restrictive LTV limits lead to high assets liquidity cost ratio, which, weakens the collateral effect, particularly in emerging market economies. Policymakers are suggested to enhance financial development and reduce liquidation frictions to enhance collateral channel in EMEs.

Keywords: Monetary Policy Transmission, Collateral Channel, Development

*We thank Dr.Ramin Nassehi for supporting research skills development, Dr.Marta Morazzoni for providing advice on modelling and Prof.Ian Preston for inspiring the introduction of precautionary saving into our model.

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1 Introduction

Since 2020, the "Great Tightening" in response to surging inflation triggers rich discussion in the effectiveness of Monetary Policy. Despite rich literature focusing on various channels , less is discussed on collateral channel that operates through changing asset values to influence consumption (Chaney et al, 2012)[5]; (Eklou, 2023)[9]. This raised our interest to investigate its effectiveness, particularly in Emerging Market Economies (EMEs)(World Economic Outlook, April 2024)[14].

This paper decompose credit constraints in collateral channel, and argues that the collateral effect of monetary policy remains minor in emerging markets with restricted credit availability, leading to little response in consumption to changing asset price. This conclusion is evident in a cross-country comparison of Colombia and Germany.

To investigate this, we adjusted Permanent Income Hypothesis model with a detailed breakdown of the credit constraint into four important parameters: economic uncertainty, LTV ratio, financial development index, and liquidation cost ratio. Cross-country comparative analysis of Germany and Colombia were conducted as an example.

The remainder of this paper is structured as follows: Section 2 reviews the existing literature on collateral channel as monetary policy transmission mechanism in EMEs; Section 3 introduces adjusted PIH model. Section 4 presents our theoretical predictions with case study. Section 5 concludes our findings.

2 Literature Review

Monetary policy transmission in EMEs is unique in its structural and institutional constraints compared to that in developed economies. Kamin and Turner (1998)[11] conclude that MPT in EMEs could be weaken for its underdeveloped financial markets.

Recent growing studies on collateral channel also supports this view. Collateral channels were found to be hindered by underdeveloped financial markets with high frictions due

to a lack of non-bank financing (Chaney et al, 2014)[5]; (European Commission, 2024)[7]. Consumers were restricted to less collateralizable assets while spending was highly leveraged (Bahaj et al., 2022)[1]; (Choi, Willems, and Yoo, 2024)[6]. Moreover, the response of housing prices was found minor in EMEs (Tunc and Gunes, 2022)[15]. This irresponsiveness was also found in credit demand to policy rate changes by Khaw and Sivabalanno (2016)[12]. This aligns with IMF's reports that the transmission through collateral channel has weakened in some countries (World Economic Outlook, Oct 2024)[8].

Some related literature suggests that macro-prudential policies, namely Loan-to-Value limits, also influence collateral channel(World Economic Outlook, April 2024)[14]. Loosening LTV limit releases credit demand when house price changes(Richter, et al, 2018)[13]. The countercyclical ability of the Optimal LTV limit was suggested to stimulate credit demand and hence collateralized spending (Ferrero et al, 2022)[10].

3 Theory

3.1 Permanent Income Hypothesis

We consider consumption to be a key intermediary reflecting the effectiveness of collateral channel. For this reason, we use the PIH model for modelling individual's consumption decisions.

The original PIH model is based on the following constrained optimisation (Carlin & Soskice, 2024)[3]:

$$\max_{C_t} U_t = \sum_{i=0}^{\infty} \frac{1}{(1+\rho)^i} \log C^e_{t+i} \tag{1}$$

$$\text{subject to: } \sum_{i=0}^{\infty} \frac{1}{(1+r)^i} C_{t+i} = (1+r)a_{t-1} + \sum_{i=0}^{\infty} \frac{1}{(1+r)^i} y^e_{t+i} = \phi_t \tag{2}$$

where ρ is the subjective discount factor, C^e_{t+i}, y^e_{t+i} represents expected consumption and expected income in the future period $t+i$, a_{t-1} represents the accumulated asset at time

$t-1$, r represents the interest rate and ϕ_t is the discounted present value of the lifetime income.

We assume our adjusted model also reflects a rational agent with forward-looking behaviour and all other agents follow the same behaviour pattern of the representative agent.

3.2 Adjusted Model

3.2.1 Asset Liquidity Cost Ratio and Interest rates

Collateral is the creditor's warranty to get the loan back. It acts as a way of exchanging fixed assets for liquid assets (**See Appendix.A**). Under PIH, through collateralising people can convert part or all of their fixed assets into liquid assets which can be used to smooth consumption. In original model 1 & 2, accumulated assets are considered to be all liquid for consumption smoothing. Therefore, we adjust PIH to include the liquidity of assets. In adjusted model, assets are categorised into fixed assets and liquid assets:

$$a_{t-1} = a_{t-1}^{\text{fixed}} + a_{t-1}^{\text{liquid}} \tag{3}$$

Fixed assets can generate rents or equivalent incomes, but there will be a conversion lag between posting a property on the market and receiving cash payments. This lag would incur a loss of interest as fixed assets cannot be compounded by interest rates. Also, changes in fixed asset valuation due to deterministic trends or interest rate fluctuation can influence fixed asset liquidity through demand and supply. These factors can exert impact on the liquidation cost ratio μ, where liquidation cost ϵ is defined as:

$$\epsilon_t = \mu_t a_{t-1}^{\text{fixed}}, \forall\ 0 \leq \mu \leq 1 \tag{4}$$

We assume that the process of collateralisation is equivalent to the liquidation of fixed

assets (**See Appendix.A**). Hence, liquidity adjusted constraint 2 is:

$$\sum_{i=0}^{\infty} \frac{1}{(1+r)^i} C_{t+i} = (1+r)(1-\mu_t) a_{t-1}^{\text{fixed}} + (1+r) a_{t-1}^{\text{liquid}} + \sum_{i=0}^{\infty} \frac{1}{(1+r)^i} y_{t+i}^{e} = \phi_t^{'} \tag{5}$$

This implies that consumption is less responsive to changes in asset valuation when taking liquidity into account (**See Appendix.B**):

$$\Delta C_t^{e'} \leq \Delta C_t^{e}, \forall\ 0 \leq \mu \leq 1 \tag{6}$$

To conclude, the higher the liquidation cost ratio μ, the lower consumptions are in response to asset value changes.

3.2.2 Economic Uncertainty

Liquidation cost ratio μ varies with economic uncertainty.

Uncertainty incurs the risk of fluctuation in asset value, which leads to the possibility of future constraints under Loan-to-Value limits. Unconstrained economic agents will precautionarily save against future credit constraints (Carroll & Kimball, 2001)[4]. Since:

$$C_t^{e} = \phi_t^{e} - S_t^{e} \tag{7}$$

optimisation problem 1 remains maximising utility as a function of consumption whilst constraint changes to the following **(See Appendix C)**:

$$\sum_{i=0}^{\infty} \frac{1}{(1+r)^i} C_{t+i} = (1+r)(1-\mu_t) a_{t-1}^{\text{fixed}} + (1+r) a_{t-1}^{\text{liquid}} + \sum_{i=0}^{\infty} \frac{1}{(1+r)^i} (y_{t+i}^{e} - S_t^{e}) = \phi_t^{'} \tag{8}$$

Uncertainty incurred precautionary saving implies less consumption for any time. Under uncertainty, reduced consumption extends the conversion lag and dampens demands for fixed assets, raising liquidation cost ϵ. Therefore, liquidation cost ratio μ is positively correlated with economic uncertainty. We consequently introduce level economic uncertainty γ in our

model such that **(See Appendix C)**:

$$\frac{\partial \mu_t}{\partial \gamma_t} > 0 \tag{9}$$

where a higher uncertainty level raises the liquidation cost ratio, causing lower responsiveness of consumption when interest rates change.

3.2.3 Financial Development Index

The Financial Development Index I^{FD} reflects the level of development of a country's financial institutions and market. The depth, access and efficiency of financial institutions are relevant to our discussion as they determine how easily consumers can liquidise their assets and how responsive asset prices react to interest rates. Emerging economies generally have lower I^{FD} compared to developed economies. Such differences in financial development will diverge consumption behaviour in developed economies and EMEs accordingly.

Higher level of financial development implies more efficient collateral channel. In economies with higher I^{FD}, consumers can collateralise fixed assets in more ways and channels. More access to financial institutions helps prices better reflect actual market demand and supplies, reducing conversion lag. Reduced lending-deposit spread would also reduce the liquidation cost **(See Appendix.A)**. Therefore, liquidation cost ratio μ is lower under higher level of financial development:

$$\frac{\partial \mu_t}{\partial I_t^{FD}} < 0 \tag{10}$$

3.2.4 Loan-to-Value(LTV) Ratio

LTV ratio θ also plays an important role in the collateral channel. It dampens consumption by freezing assets. If θ <1 (which is usually the case), part of the value of fixed assets is not

liquidated:

$$\text{frozen assets} = (1-\theta_t)a_{t-1}^{\text{fixed}} \tag{11}$$

Accordingly, the adjusted constraint is:

$$\sum_{i=0}^{\infty}\frac{1}{(1+r)^i}C_{t+i} = (1+r)(1-\mu_t)\theta a_{t-1}^{\text{fixed}} + (1+r)a_{t-1}^{\text{liquid}} + \sum_{i=0}^{\infty}\frac{1}{(1+r)^i}y_{t+i}^{e} - S_{t+i} = \phi_t^{'} \tag{12}$$

which implies that:

$$\frac{\partial C_t}{\partial \theta_t} > 0 \tag{13}$$

This incomplete liquidation also implies that consumption response to interest rate changes is asymmetric. Consumption is less responsive to interest rate cuts as frozen assets are unavailable for consumption smoothing, while it is responsive to increasing interest rates as liquid assets are not influenced. This might also influenced by uncertainty and LTV limit **(See Appendix D)**. The asymmetry might explain resilient saving in EMEs where interest rate cuts would not trigger consumptions as expected.

4 Results

4.1 Consumption Function

Using the theoretical framework above, we define a consumption function as:

$$C_t := f(\mu_t, \gamma_t, \theta_t, I_t^{FD}) \tag{14}$$

where μ is the liquidation cost ratio, γ is the economic uncertainty, θ is LTV ratio and I^{FD} is financial development index, all at time t. From theory framework, we have:

$$\frac{\partial C_t}{\partial \mu_t} < 0, \frac{\partial C_t}{\partial \theta_t} > 0 \tag{15}$$

From 9 and 10 **(See Appendix E)**, we have :

$$\frac{\partial C_t}{\partial \gamma_t} < 0, \frac{\partial C_t}{\partial I_t^{FD}} > 0 \tag{16}$$

Through this consumption function, we investigate how characteristics of the four parameters in EMEs can impact its monetary policy transmission.

4.2 Theoretical Predictions in EMEs

As shown in Figure 1, through the collateral channel, the liquidated value of fixed assets is determined by both the average LTV ratio of the economy and the liquidation cost ratio. Different characteristics of economic uncertainty and financial developments in EMEs and developed economies diverge their respective consumption response.

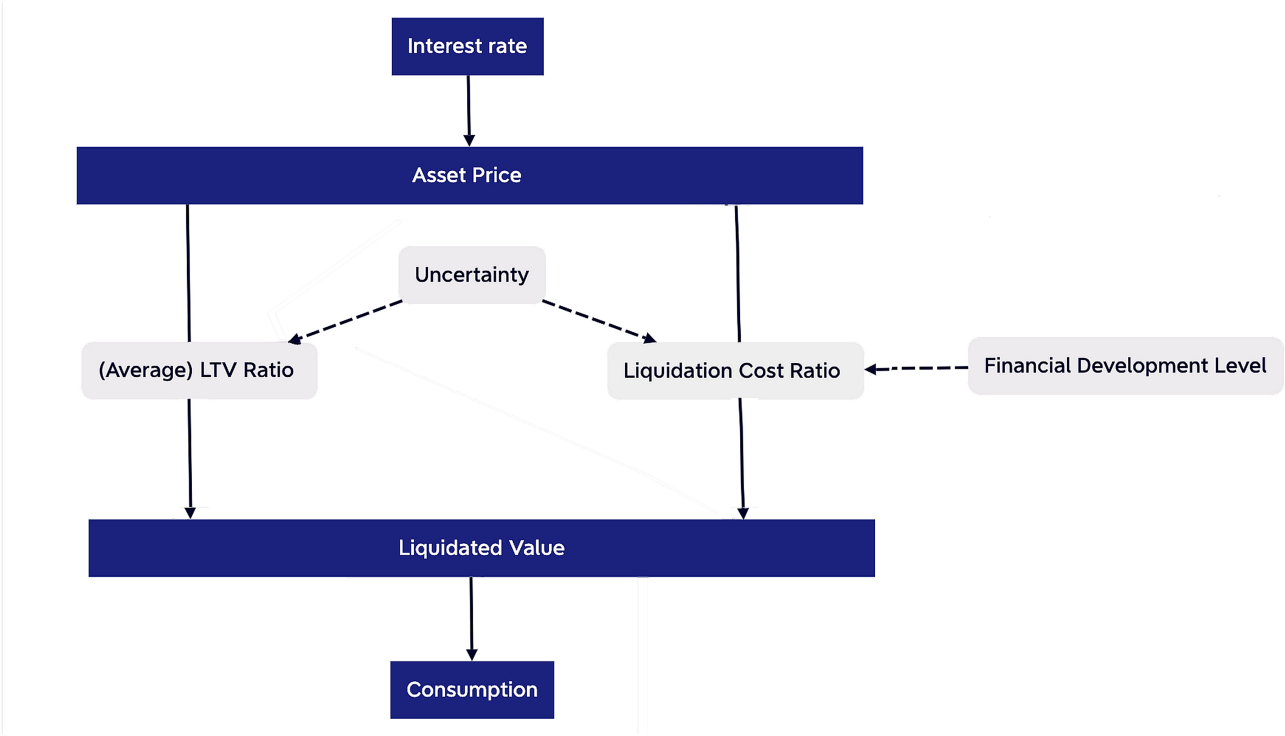


Figure 1: Collateral Channel

In EMEs, the level of economic uncertainty γ is higher (Bloom, 2014)[2] and financial development I^{FD} is lower compared to developed economies. With these two characteristics, the liquidation cost ratio μ of fixed assets is higher, causing lower consumption adjustment in EMEs under the same interest rate changes. Meanwhile, LTV ratio of EMEs is lower as consumers are less willing to smooth their consumption under higher uncertainty, presenting hand-to-mouth behaviour (Carlin & Soskice, 2024)[3]. The resultant consumption change in EMEs in response to interest rate adjustments is lower.

Thus, ceteris paribus, collateral channel in EMEs is weakened due to its characteristics in economic uncertainty and financial development.

4.3 Colombia-Germany Comparision

The comparison of Germany and Colombia provides a strong empirical validation for our predictions.

Country	FDI (I_t^{FD})	CDSS (γ)	LTV (θ)	LTV Limit (θ^{lim})
Colombia	0.39	149.16	53%	80%
Germany	0.71	15.7	82%	Infinite

Table 1: Financial Indicators for Hungary and Germany

As listed in Table 1, Germany has a highly developed financial market, with a high I_t^{FD}[1] of 0.71 while Columbia has only 0.39. Their difference in θ[2] is also significant. Colombia imposes 80% θ^{lim}, while its average θ is only 53%, signalling higher uncertainty-induced precautionary saving. Germany has no official θ^{lim} and an average 82% θ, represented by a high indebtedness in data (Fig 2).

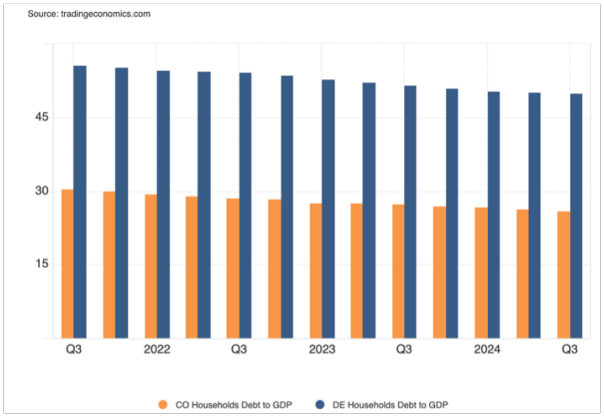


Figure 2: Household Debt to GDP

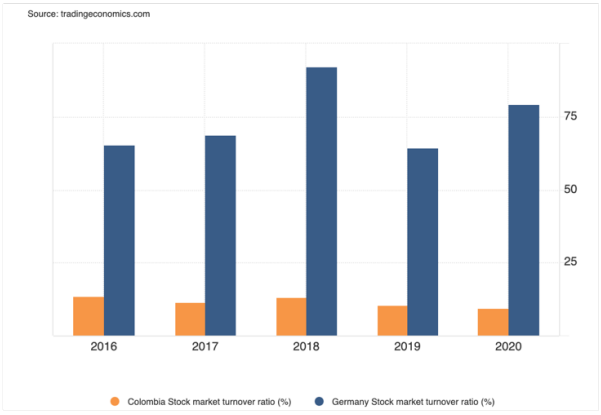


Figure 3: Stockmarket Turnover Rates

Despite no direct measure of asset liquidity cost ratio exists, Germany's larger mortgage market and higher transaction efficiency suggest lower liquidation costs with a 79% of stock market turnover rate, compared to only 9% in Colombia (Fig 3), reflecting an inactive market and higher liquidation cost. Uncertainty is measured by Credit Default Swap Spreads

[1]Financial Development Index: Financial Development - IMF Data (2025) Financial Development Index. https://data.imf.org/?sk=F8032E80-B36C-43B1-AC26-493C5B1CD33B&sId=1485894037365

[2]Loan-To Value Ratio & Limit: IMF reports of Colombia and Germany.(IMF, 2017; 'Germany IMF Report'

(CDSS)[3]. Compared to an index of 15.7 in Germany, it is shockingly high as 149.16 in Colombia, reflecting concerns of default.

Based on our theoretical predictions, colombia's consumption change in response to house price would be less significant, as they are more credit constrained compared to Germany. The data showed in Fig 4.1 (Colombia) and Fig 4.2 (Germany) supports this conclusion. In Colombia, a weak collateral channel is suggested as housing price and consumer spending show a great divergence in trend.



Figure 4: Percentage Change in Housing Price and Consumer Spending in Colombia

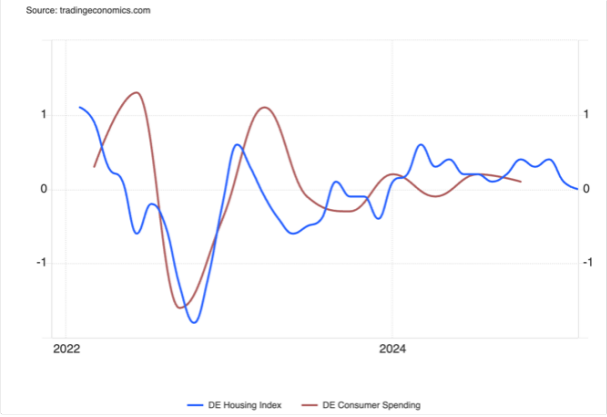


Figure 5: Percentage Change in Housing Price and Consumer Spending in Germany

In contrast, Germany's consumer spending shows responsiveness to house price change. More flexibility in collateralization through loosening LTV limit and reduced liquidation cost would enhance consumer's ability to adjust their consumption with asset price change. This Cross-Country comparative statics emphasizes the role of the four parameters $(I^{FD}, \gamma, \mu, \theta)$ on the effectiveness of collateral channel.

[3]Credit Default Swaps Spread: GovBonds, M. (2023) CDs historical data, World Government Bonds. Available at: https://www.worldgovernmentbonds.com/cds-historical-data/. A credit default swap (CDS) is a financial derivative that allows an investor to swap or offset their credit risk with that of another investor. In other words, The CDS Spread is the price of a credit default swap. Higher sovereign CDS spreads indicate greater perceived credit risk and economic uncertainty.

5 Key Findings & Policy Implications (Conclusion)

The effectiveness of the collateral channel is largely determined by credit constraints through the liquidity cost ratio and the LTV ratio. In emerging economies, higher liquidation cost ratio is driven by low I^{FD} in inactive asset markets, while economic uncertainty tightens credit constraints by inducing precautionary saving, reducing LTV ratio. Thus, the amplified restriction in credit leads to irresponsiveness of consumption to asset price change.

We suggest that policies aimed at reducing liquidation frictions, enhancing credit market efficiency, and mitigating excessive risk aversion can improve the effectiveness of collateral channel in emerging economies.

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Appendix.A

By collateralising, fixed assets are no longer part of the asset that is available for consumption smoothing. Before the loan is repaid, the term reflecting fixed assets in the constraint is [4]:

$$(1+r_t)(1-\mu_t)\theta_t a_{t-1}^{\text{fixed}} = (1+r_t)a_{t-1}^{\text{liquidated}} \tag{17}$$

As consumers can use liquidated assets to smooth consumption, the liquidated asset term can be expressed as the infinite sum of liquidated value at each period, each term is necessarily the same to remain utility maximisation:

$$(1+r_t)a_{t-1}^{\text{liquidated}} = \sum_{i=0}^{\infty} \frac{1}{(1+r)^i} m_{t+i} \tag{18}$$

The infinite sum of instalments of loan that should be repaid at each period is the same as the liquidated value, and each term is the same:

$$\sum_{i=0}^{\infty} \frac{1}{(1+r)^i} m_{t+i} = \sum_{i=0}^{\infty} \frac{1}{(1+r)^i} l_{t+i} \iff \frac{1+r}{r} m_{t+i} = \frac{1+r}{r} l_{t+i} \iff m_{t+i} = l_{t+i} \tag{19}$$

Substituting back to the budget constraint:

$$\sum_{i=0}^{\infty} \frac{1}{(1+r)^i} C_{t+i} = (1+r)a_{t-1}^{\text{liquid}} + \sum_{i=0}^{\infty} \frac{1}{(1+r)^i} (y_{t+i}^e - S_{t+i} + m_{t+i} - l_{t+i}) \tag{20}$$

m_{t+i} and l_{t+i} in 20 seem naturally cancel each other out, representing that consumers still claim the ownership of the collateralised fixed asset.

Realistically, in the existence of lending-deposit spread, lending rate is higher than deposit rate. Higher level of financial development would reduce this spread. Assuming that r_t

[4] θ is the LTV ratio which was introduced in later sections

remains the same and $r_t < r^{\text{lending}}$, define $f(r) = \frac{1+r}{r}$ of which :

$$f'(r) = -\frac{1}{r^2} > 0, \forall r > 0 \tag{21}$$

meaning that increase in r would lead to decrease in $f(r)$. Since $0 < r_t < r^{\text{lending}}$:

$$\sum_{i=0}^{\infty} \frac{1}{(1+r_t^{\text{lending}})^i} l_{t+i} = \frac{1+r_t^{\text{lending}}}{r_t^{\text{lending}}} < \frac{1+r_t}{r_t} = \sum_{i=0}^{\infty} \frac{1}{(1+r_t)^i} l_{t+i} \tag{22}$$

Recall the definition of liquidation cost:

$$\epsilon_t = \mu_t \theta_t (1+r_t) a^{\text{fixed}} = \theta_t (1+r_t) a^{\text{fixed}} - \sum_{i=0}^{\infty} \frac{1}{(1+r_t)^i} l_{t+i} \tag{23}$$

Under lending-deposit spread, liquidation cost is higher:

$$\epsilon_t^{\text{with spread}} = \theta_t (1+r_t) a^{\text{fixed}} - \sum_{i=0}^{\infty} \frac{1}{(1+r_t^{\text{lending}})^i} l_{t+i} > \epsilon_t \tag{24}$$

This result implies that liquidation cost ratio μ is increased in the existence of lending-deposit spread, which supports:

$$\frac{\partial \mu_t}{\partial I_t^{FD}} = \underbrace{\frac{\partial \mu_t}{\partial \text{ lending-deposit spread}}}_{+} \underbrace{\frac{\partial \text{ lending-deposit spread}}{\partial I_t^{FD}}}_{-} < 0 \tag{25}$$

Appendix.B

For this optimisation problem 1 & 2:

Marginal utility at time t should be equal to marginal utility at time t+1 for optimising utility across all periods:

$$\frac{1}{c_t} = \frac{1+r}{1+\rho}\frac{1}{C^e_{t+1}} \tag{26}$$

By assuming the subjective discount rate is equal to the interest rate:

$$C_t = C^e_{t+1} = C^e_{t+i} \tag{27}$$

Substitute back to the constraint we have:

$$C_t \sum_{i=0}^{\infty} \frac{1}{(1+r)^i} = \phi_t \tag{28}$$

By infinite sum of geometric progression:

$$C_t \frac{1}{(1-\frac{1}{1+r})} = \phi_t \implies C_t = \frac{r}{1+r}\phi_t \tag{29}$$

An increase in interest rates reduces the present value of fixed assets. So asset value has a negative correlation with changes in interest rates. Other things being equal, changes in asset value are the result of adjustments in interest rates. The effect of change in asset value in consumption is :

$$\Delta C^{e\prime}_t = \frac{r}{1+r}(1-\mu_t)(1+r)\Delta a^{\text{fixed}}_{t-1} = r(1-\mu_t)\Delta a^{\text{fixed}}_{t-1} \tag{30}$$

which is strictly smaller than the effect on consumption due to change in asset valuation in the unadjusted model for all $\mu_t < 1$:

$$\Delta C^{e}_t = \frac{r}{1+r}(1+r)\Delta a^{\text{fixed}}_{t-1} = r\Delta a^{\text{fixed}}_{t-1} \tag{31}$$

Appendix.C

Consumption, saving and uncertainty

By Appendix.B and PIH model, we have:

$$C_t = (1-\gamma)\frac{r}{1+r}\phi_t^e \tag{32}$$

$$S_t = \gamma\frac{r}{1+r}\phi_t^e \tag{33}$$

$$C_t + S_t = \frac{r}{1+r}\phi_t^e \iff \sum_{i=0}^{\infty}\frac{1}{(1+r)^i}(C_{t+i}^e + S_{t+i}^e) = \phi_t^e \tag{34}$$

From this we have:

$$\frac{\partial S_t^e}{\partial \gamma_t} > 0 \tag{35}$$

We can obtain from the definition of liquidation cost that, with higher consumption, more demand is in the market that reduces conversion lag:

$$\frac{\partial \mu_t}{\partial C_t^e} < 0 \tag{36}$$

Also from $C_t = \phi_t - S_t$:

$$\frac{\partial C_t^e}{\partial S_t^e} < 0 \tag{37}$$

Rearranging the partial derivatives by chain rule:

$$\frac{\partial \mu_t}{\partial \gamma_t} = \underset{-}{\frac{\partial \mu_t}{\partial C_t^e}}\underset{-}{\frac{\partial C_t^e}{\partial S_t^e}}\underset{+}{\frac{\partial S_t^e}{\partial \gamma_t}} > 0 \tag{38}$$

Appendix.D

The Role of LTV Limits

θ comes with a limit to avoid systemic risk to the financial system. The lower the LTV limit is, the less agents can smooth consumption as begin more credit-constrained. Due to different levels of regulation, LTV limit θ^{lim} varies across countries. The difference between θ^{lim} and θ reflects consumer's risk aversion behaviour, which is correlated with the economic uncertainty mentioned above:

$$\frac{\partial(\theta_t^{lim} - \theta_t)}{\partial \gamma_t} > 0 \tag{39}$$

The more uncertain the economy is, the more risk averse consumers are:

$$\frac{\partial \theta_t}{\partial \gamma_t} < 0 \tag{40}$$

which means that consumers will self-constrain to be invariant to changes in interest rates when the economic uncertainty is rising. This is the collateral channel emulation of precautionary saving.

Appendix.E

Use all the result derived from Appendix.A and Appendix.C, we obtain:

$$\frac{\partial C_t}{\partial \gamma_t} = \underset{-}{\frac{\partial C_t}{\partial \mu_t}} \underset{+}{\frac{\partial \mu_t}{\partial \gamma_t}} < 0 \tag{41}$$

$$\frac{\partial C_t}{\partial I_t^{FD}} = \underset{-}{\frac{\partial C_t}{\partial \mu_t}} \underset{-}{\frac{\partial \mu_t}{\partial I_t^{FD}}} > 0 \tag{42}$$